Final Project Notebook

DS 5001 Text as Data

Metadata

- Full Name: Donald Williams
- Nick Name: dw
- GitHub Repo URL: https://github.com/piedmontdata/DS5_Final_project
- Tableau URL: https://public.tableau.com/profile/donald5297#!/vizhome/DS5_Final_Project/Story1

Overview

The goal of this project for me is to see if I can create a predictive model of a movie using the tone, structure, and perspective, we're learning in this course. I'm not sure if a movie that has deeply been digitized and transcribed, seen that has sit in computer set of tables, and then generated and visualize the results of a series of fitted models. I'm still also draw some tentative conclusions regarding the linguistic, structure, and tone of movie that I'm interested in watching based on the results of the models. I'm also pair with a colleague through the entire pipeline from beginning to conclusion.

Specifically, you will complete a variation of the following tasks:

- **Dataset** Do collection from their source formats (TXT) into a set of tables conforming to the standard Text Analysis Data Model (TAD).
- **Analytic** Make some descriptive and inferential and hypothesis tests using that tabular data and NLP (TA).
- **Model** Fit a model to the data, using the tabular data and NLP features derived from the text (TM).
- **Model** Do a provided and associated model with tables and features derived from the application of unsupervised methods, including PCA, LDA, and clustering. (TM).
- **Model** Do a provided and associated model with tables and features derived from the application of supervised methods, including KNN, Naive Bayes, and Decision Trees. (TM).

When you are finished, you will make the results of your work available in GitHub (for code) and Tableau (for data). You will submit to Blackboard your Tableau Public URL and a zipped or tarred archive of your repository on the GitHub/Tableau Public.

Some Details

- Please ID your variables in each table below by adding the mentioned.
- Replace text that asks you to insert something with the thing, e.g. replace `CREATED_TABLE_NAME` with an imaginary table, e.g. `TITANIC.csv`.
- Do not change the table headers, e.g. `VOCAB(1)` should be `VOCAB(1)` in your completed table, e.g. `TITANIC.csv(1)`.
- Please do not alter the order of the document or, as I, or the provided text.
- The top with explanatory paragraphs belongs before the tabular data.
- Please do not alter the order of the tabular data, e.g. the vocabulary.
- Tables not indexed by tokens will cause a penalties.

Raw Data

Source Description (1)

Provide a brief description of the original raw material, including its provenance and content. Tell us where you found it and what kind of content it contains.

The primary source material that will be analyzed throughout this project is an Magazine. An magazine was a weekly published magazine from 1967 to 2014. An magazine was a journal/reading publishing company that specialized in publishing short stories, novels, and other literary works. The magazine consisted of a wide range of genres that included, science fiction, fantasy, and horror.

The analysis will be conducted on 150 issues of an magazine that range from 1967 to 1979. The magazine archives were collected from the website, Internet Archive. The process of collecting the magazines was time consuming and involved I had to manually download the issues from Internet Archive. After each magazine was downloaded in the zipped format I pulled the text files into plaintext and converted it to a pdf file.

Source Features (1)

Add details for the following items. (Do this for all following tabular files.)

- Source URL:
  - https://archive.org/details/dp3magazine_miscellaneous?sort%5B0%5D=%5B%5D=mediatype%3A%28texts%29
- Link to collection:
  - https://archive.org/details/dp3magazine?sort%5B0%5D=%5B%5D=mediatype%3A%28texts%29
- Number of documents:
  - 150
- File format:
  - Text, plain
- Time span of your documents (e.g. 1789):
  - 1967-1979
- Number of words (e.g. 1789), unlabeled, etc.:
  - 2,000,000

Source Document Structure (1)

Provide a brief description of the structure of your documents. Tell us how they are organized. Describe the logical structure in structural terms relative to each other. For example, a corpus of letters might be described as having a date, an addressee, a salutation, a set of content paragraphs, and closing. If they are without structure, state that.

The internal structure of the magazine issues will vary from issue to issue. Each issue contains stories, cartoons, and ads. Most of the ads I magazine documents, an magazine focused on fashion, beauty and also contained a section that was devoted to cartoons.

Each text primary consists of headline, chapter/section/subsection, or a three-body paragraphs, and a concluding paragraph. These text primary are usually accompanied by images. Each text primary is consecutive to its neighbors, there is sometimes a text primary that will continue on to 3 or 4 pages through out the magazine issue.

Unfortunately, due to how messy the data was in it difficult to break up the text by its whole and which headline it belongs under. Therefore the OHCO structure for the analysis will be reduced to the magazine issue level and not the text primary level.

Parsed and Annotated Data

Place the raw files into the files sub-folder of your collection. Use `CORPUS`, and `VOCAB` names.

These tables will be stored as CSV files with header rows.

Use the variable name `p` for paragraphs.

Provide the following information for each.

LIB (2)

A table which documents the corpus composition. These may be books, plays, newspaper articles, abstracts, blog posts, etc.

- GitHub URL for raw(ish) source:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/LIB.csv
- Link to URL for readable source(s):
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/LIB.csv
- Number:
  - 150
- Number of observations:
  - 150
- Type of documents, including at least those that may be used for model construction (e.g. data, outlier, etc.):
  - Science fiction
  - Quarter published
  - Contains short stories and novels
  - Short stories
  - Novels
  - Cartoons
  - Ads
- Table of contents:
  - `doc_id`
  - `title`
  - `year_published`
  - `issue_number`

CORPUS (2)

The sequence of word tokens in the corpus, indexed by their location in the corpus and document structures.

- GitHub URL:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/CORPUS.zip
- GitHub URL for notebook used to create:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- Number:
  - 2,000,000
- Number of observations (between (chunk/line or 100,000 and 1,000,000 observations):
  - 2,000,000
- Add'l features (e.g. parts-of-speech, named entities):
  - `stop_id`, represents the magazine id
  - `part_of_speech`, represents the part of speech
  - `token_num`, represents the token number
- Table of contents (paste below):
  - `doc_id`
  - `token_num`
  - `token_id`
  - `token_count`
  - `token`
  - `stop_id`
  - `part_of_speech`
  - `date_time_digitized`

VOCAB (2)

The unique word types (lexica) in the corpus.

- GitHub URL:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/VOCAB.zip
- GitHub URL for notebook used to create:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- Number:
  - 50,000
- Number of observations:
  - 50,000
- Table of contents:
  - `token`
  - `porterstem`
  - `pos_tuple`
  - `definitions`
  - `synonyms`
  - `hypernyms`
  - `hyponyms`
  - `antonyms`
  - `derivationally_related_forms`
  - `lemma_names`
  - `verb_frames`
  - `lexnames`
  - `min_depth`
  - `max_depth`
  - `frequency`
  - `stopword`
  - `contraction`
  - `isalpha`
  - `isupper`
  - `islower`
  - `lemma`
  - `stem`
  - `pos`
  - `sentiment`

Relational Design

token_id	n_chars	p	l	max_pos	max_pos_group	stop	stem_porter	dfidf
shell	5	0.000041	14.202255	NNS	NN	0	shell	5.267710
springfield	4	11	0.000035	15.009610	NNP	NN	0	springfield	5.267710
squad	5	11	0.000035	14.202255	NN	NN	0	squad	5.267710
demonstrate	4	11	0.000035	15.009610	NN	NN	0	demonstr	5.267710
demands	5	7	0.000031	14.567663	NNS	NN	0	demand	5.267710
delivery	7	7	0.000031	14.202255	NNP	NN	0	deliver	5.267710
delivery	7	8	0.000031	14.202255	NN	NN	0	deliver	5.267710
stabbed	6	7	0.000031	14.434647	VBD	VB	0	stab	5.267710
delicate	4	8	0.000035	15.009610	JJ	JJ	0	delic	5.267710
stadium	6	8	0.000035	14.567663	NNP	NN	0	stadium	5.267710
detaware	6	8	0.000041	14.434647	NNP	NN	0	detaware	5.267710
detangle	6	8	0.000035	15.009610	NNP	NN	0	detangl	5.267710
definitely	4	10	0.000035	15.009610	RB	RB	0	defint	5.267710
standard	5	10	0.000041	14.434647	NNP	NN	0	standard	5.267710
defendant	6	10	0.000041	14.434647	NN	NN	0	defend	5.267710
detroit	6	10	0.000041	15.009610	NNP	NN	0	detroit	5.267710
defeat	7	6	0.000031	14.202255	NN	NN	0	defeat	5.267710
dedicated	9	6	0.000041	14.434647	VBN	VB	0	dedic	5.267710
decided	7	6	0.000031	15.009610	VBD	VB	0	decidn	5.267710
starred	4	7	0.000031	15.009610	VBD	VB	0	star	5.267710

Derived Tables

BOW (3)

A bag-of-words representation of the CORPUS.

- GitHub URL:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/BOW.zip
- GitHub URL for notebook used to create:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- Number:
  - 150
- Key (represents terms of bow):
  - `doc_id`
- Number of observations:
  - 150
- Key (represents terms of bow), including `n_OTHER`:
  - `doc_id`

DTM (3)

A representation of the BOW as a sparse count matrix.

- GitHub URL:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/DTM.zip
- GitHub URL of BOW used to generate (if applicable):
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- GitHub URL for notebook used to create:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- Number:
  - 150
- Key (represents terms of DTM):
  - `doc_id`
- Number of observations:
  - 150

TFIDF (3)

A Document Term matrix with TFIDF values.

- GitHub URL:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/TFIDF.zip
- GitHub URL of BOW used to generate (if applicable):
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- GitHub URL for notebook used to create:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- Number:
  - 150
- Key (represents terms of DTM or TFIDF):
  - `doc_id`
- Description of TFIDF Errors (`dfIDF` 0/N):
  - `dfIDF` 0/N = `dfIDF` / `n_docs`
  - If this document has terms in vocabulary (`n_OTHER` is by code):
    - 0 = target all OTHER
    - 1 = all vocabulary represented

Reduced and Normalized TFIDF_L2 (3)

A Document Term matrix with L2-normalized TFIDF values.

- GitHub URL:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Data/Raw%20Data/TFIDF_L2.zip
- GitHub URL of BOW used to generate (if applicable):
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- GitHub URL for notebook used to create:
  - https://github.com/piedmontdata/DS5_Final_project/blob/master/Code/Parsing%20and%20Tokenization%20Notebook.ipynb
- Number:
  - 150
- Number of features (i.e. significant vocabs):
  - 50,000
- Key (represents terms of TFIDF_L2):
  - The threshold used was 0.25. The original quality of this data was very poor so I decided to increase the threshold to an attempt to filter out some of the noisy, condensed stories that I couldn't fully remove from the Porter Stemmed Lemmatized Data. Shall I be' replacement the choice of each word, my children were like to incorporate the thermostat I could remove control control control

There doesn't seem to be a clear pattern between these two principal components and the mean document weight of each topic.

Sentiment VOCAB_SENT (4)

Sentiment values associated with the VOCAB from a curated sentiment lexicon.

- LIVA Box URL:
  - https://virginia.box.com/s/1ntrjvonvnwk27ekje5i0ypezk1k44
- LIVA Box URL for source lexicon:
  - https://virginia.box.com/s/2d7fm7f23uq5h1z24ua6mx3q7h2xm
- GitHub URL for notebook used to create:
  - https://github.com/jenyalielas/ds5001_final_project/blob/main/sentiment_analysis.ipynb
- Delimiter:
  - ","

Sentiment BOW_SENT (4)

Sentiment values from VOCAB_SENT mapped onto BOW.

- LIVA Box URL:
  - https://virginia.box.com/s/7t2xqbqcwxqo0cqdxcr3qgbqi2kthaj
- GitHub URL for notebook used to create:
  - https://github.com/jenyalielas/ds5001_final_project/blob/main/sentiment_analysis.ipynb
- Delimiter:
  - ","

Sentiment DOC_SENT (4)

Computed sentiment per bag computed from BOW_SENT.

- LIVA Box URL:
  - https://virginia.box.com/s/ur5ifxhoo5f6fhmwb5dvrq9ysvln0qfcq
- GitHub URL for notebook used to create:
  - https://github.com/jenyalielas/ds5001_final_project/blob/main/sentiment_analysis.ipynb
- Delimiter:
  - ","
- Document bag expressed in terms of OHCO levels:
  - "mag_id"

Sentiment Plot (4)

Plot sentiment over some metric space, such as time.

If you don't have a metric metadata features, plot sentiment over a feature of your choice.

You may use a bar chart or a line graph.

The previous image the mean and overall sentiment of the magazines plotted over the years that they were released. It seems that both the overall and mean sentiment have increased over time. However, the mean sentiment has only increased on a very small scale.

VOCAB_W2V (4)

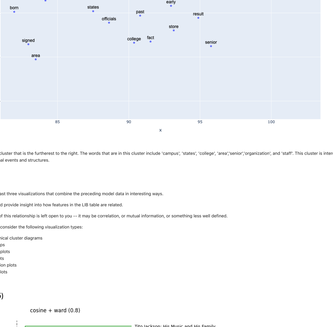
A table of word2vec features associated with terms in the VOCAB table.

- LIVA Box URL:
  - https://virginia.box.com/s/sfxfd9qc6ltzxkopm2xfxw7z9wsyraxd2e
- GitHub URL for notebook used to create:
  - https://github.com/jenyalielas/ds5001_final_project/blob/main/word2vec.ipynb
- Delimiter:
  - ","
- Document bag expressed in terms of OHCO levels:
  - "mag_id"
- Number of features generated:
  - 100
- The library used to generate the embeddings:
  - gensim

Word2vec tSNE Plot (4)

Plot word embedding features in two-dimensions using t-SNE.

Describe a cluster in the plot that captures your attention.



I chose the cluster that is the furthered to the right. The words that are in this cluster include 'campus', 'states', 'college', 'area','senior','organization', and 'staff'. This cluster is interesting because it relates to organizational events and structures.

Riffs

Provide at least three visualizations that combine the preceding model data in interesting ways.

These should provide insight into how features in the LIB table are related.

The nature of this relationship is left open to you --- it may be correlation, or mutual information, or something less well defined.

In doing so, consider the following visualization types:

- Hierarchical cluster diagrams
- Heatmaps
- Scatter plots
- KDE plots
- Dispersion plots
- t-SNE plots
- etc.

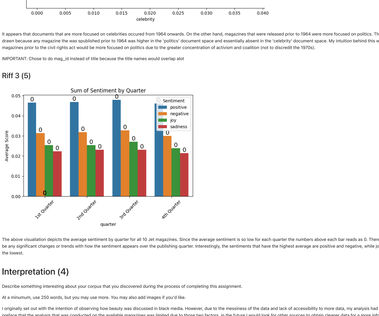
Riff 1 (5)

When looking at the dendrogram we see that the magazines are broken into two primary groups. The first primary group consists of 3 clades that results in four leaves. The four leaves are the following: 'Tito Jackson: His Music and His Family', 'What Dope Arrest is Doing to Career of Barbara McNair', 'Benson Shares His Fame With Family', and 'New Reggie Defends His Hero Image'. These four magazines are the closest to one another due to their central focus on a celebrity figure. Interestingly, the other magazine that is closest to these clades is titled, 'What's Behind Suicide of Charles Diggs Sr.?' which also focuses on a celebrity but not in the same positive light as the other magazines.

The second primary group focuses primarily on more political and survey focused information.

Overall, all the magazines seem to similar due to their heights being similar heights to one another.

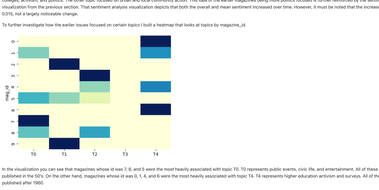
Riff 2 (5)



It appears that documents that are more focused on celebrities occured from 1964 onwards. On the other hand, magazines that were released prior to 1964 were more focused on politics. This conclusions is drawn because any magazine the was spublished prior to 1964 was higher in the 'politics' document space and essentially absent in the 'celebrity' document space. My intuition behind this was that jet magazines prior to the civil rights act would be more focused on politics due to the greater concentration of activism and coalition (not to discredit the 1970s).

IMPORTANT: Chose to do mag_id instead of title because the title names would overlap alot

Riff 3 (5)



The above visualation depicts the average sentiment by quarter for all 10 jet magazines. Since the average sentiment is so low for each quarter the numbers above each bar reads as 0. There doesn't appear to be any significant changes or trends with how the sentiment appears over the publishing quarter. Interestingly, the sentiments that have the highest average are positive and negative, while joy and sadness are the lowest.

Interpretation (4)

Describe something interesting about your corpus that you discovered during the process of completing this assignment.

At a minumum, use 250 words, but you may use more. You may also add images if you'd like.

I originally set out with the intention of observing how beauty was discussed in black media. However, due to the messiness of the data and lack of accessibility to more data, my analysis had to be shifted. I will preface that the analysis that was conducted on the available magazines was limited due to those two factors, in the future I would look for other sources to obtain cleaner data for a more informative analysis. However, throughout this analysis I was able to gain some interesting insights.

Throughout this analysis, there have been some interesting findings regarding how Jet magazine discussed black political figures, celebrities, and activism from 1951 to 1978. First as demonstrated with the first BIFF visualization, Jet magazines that were published prior to 1964 were more concentrated in the 'politics' than 'celebrity' document space. This could be due to numerous key political and social moments that occurred throughout the 1950's and 1960's. Examples include Brown v. Board of Education ruling, the Montgomery Bus Boycotts, Little Rock Nine, and the Civil Rights Act passing.These examples demonstrate that this time period was more politically driven and centered and ranged from school desegregation to economic boycotts. I believe that this is further reinforced by two of the LDA topics in which one focused on colleges, activism, and politics. The other topic focused on urban and local community action. This idea of the earlier magazines being more politics focused is further reinforced by the sentiment analysis visualization from the previous section. That sentiment analysis visualization depicts that both the overall and mean sentiment increased over time. However, it must be noted that the increase in sentiment is by 0.015, not a largely noticeable change.

To further investigate how the earlier issues focused on certain topics I built a heatmap that looks at topics by magazine_id.

In the visualization you can see that magazines whose id was 7, 8, and 5 were the most heavily associated with topic T0. T0 represents public events, civic life, and entertainment. All of these magazines were published in the 50's. On the other hand, magazines whose id was 0, 1, 4, and 6 were the most heavily associated with topic T4. T4 represents higher education activism and surveys. All of these magazines were published after 1960.

This visualization doesn't confirm the previous intuition about the shift in what Jet magazines discussed from the 1950's to the 1970's due to the later issues being most heavily associated with the activism topic. To approve upon this analysis, I would gather more magazines to account for the sparsity in data. By accounting for sparsity, analyzing the shift in what was discussed would be more intelligible and could give a clearer picture of when or if a shift occurred.